

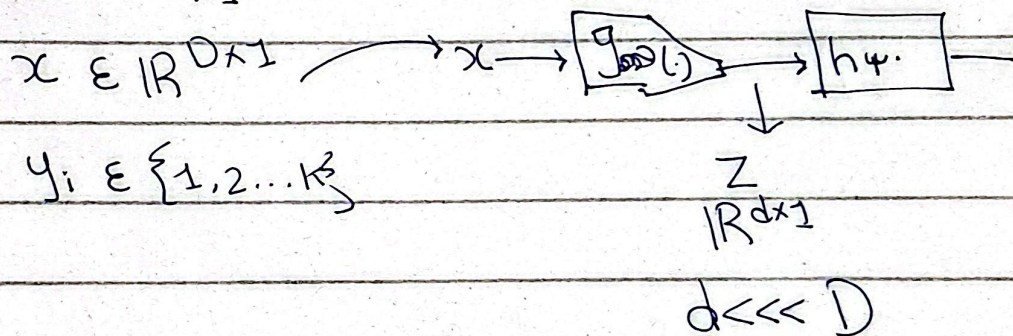
Lecture 3

$$WD_1 = \langle \pi, C \rangle \quad \text{if } C_{ij} = |i-j|$$

$$WD_2 = \langle \pi, C \rangle \quad \text{if } C_{ij} = |i-j|^2$$

* Discriminative models:

$$D = \{x_i, y_i\}_{i=1}^N$$



$$\begin{aligned}
 &\rightarrow P(x, y) \rightarrow P(y) \\
 &\rightarrow P(z) \rightarrow P(y|x) \\
 &\rightarrow P(x) \rightarrow P(y|z)
 \end{aligned}$$

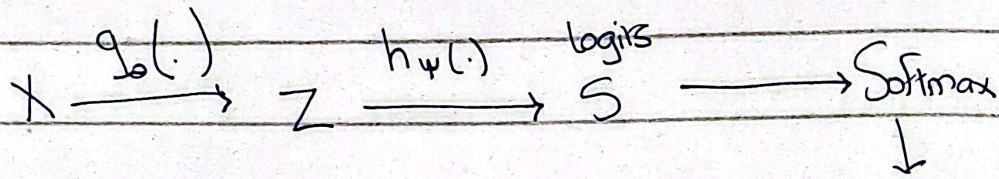
like if we have an image cat, what will be the probability of each class with respect to x.

↓
all label's probability wrt cat is distributed

$P - P_{\theta}(Y|Z)$ - model's true representation

$Q - P_{\psi}(Y|Z)$ - model's learnt representation

$$P(Y|X) \approx P(Y|Z)$$



$P \xrightarrow{\text{model's true representation}} P_{\psi}(Y|Z)$

↑



$$Y \rightarrow H(Y)$$

$$Y|X \rightarrow H(Y|X) \rightarrow \text{entropy of } Y \text{ given } X \text{ has already happened}$$

- What is the entropy of $Y|X$ compared to Y

$$I(Y, X) = H(Y) - H(Y|X)$$

$$\Rightarrow I(Y|Z) = H(Y) - H(Y|Z)$$

↓
mutual information

↓
fixed

From the given
dataset

↓
make Z
predictive of
 Y

→ we need to decrease $H(Y|Z)$,

(so $I(Y|Z)$ will be higher

in order to decrease this:

$$H(Y|Z) = - \int P(Z) \log(P$$

$$H(Y|Z) = - \int P(Y|Z) \log(P(Y|Z)) dz$$

True distribution of

Y given Z ,

which is never available

$$L_{CE} = -\log P_Y(Y=j|Z) \leftarrow \text{so we estimate using cross entropy loss}$$

Negative log Likelihood

NLL

$Y=j$ — True class

— The less $P(Y=j, Z)$ is

the higher the loss is

$$\rightarrow P(Y=j, Z)$$

for each example z , what's the prob of its true label

$$L_{CE} = \frac{1}{N} \sum_{i=1}^N -\log P_Y(Y=j, z_i)$$

$$* \overline{D_{KL}(P_Y(Y|Z) || P(Y|Z))} \geq 0$$

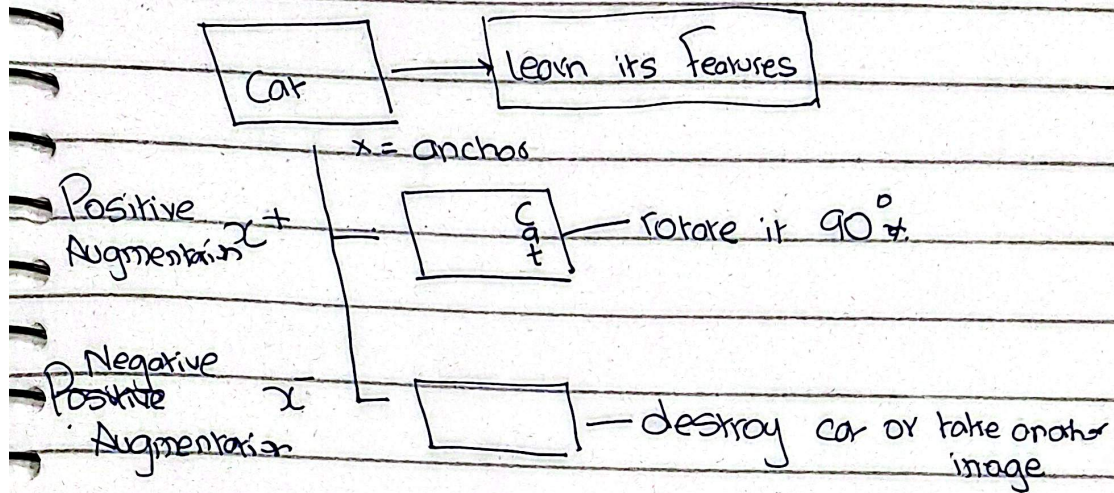
↓
0 only if $P_Y(Y|Z) = P(Y|Z)$
distance will be positive always

$$L_{CE} = -T(Y|Z) + D_{KL}(\cdot)$$

* Model trained on cross-entropy [Discriminative model]
[in which we decrease cross-entropy loss]

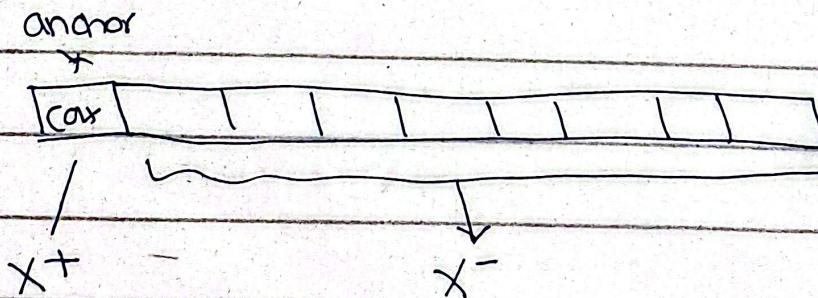
2 — Contrastive Models

$$\text{Data} = \{x_i\}_{i=1}^n$$



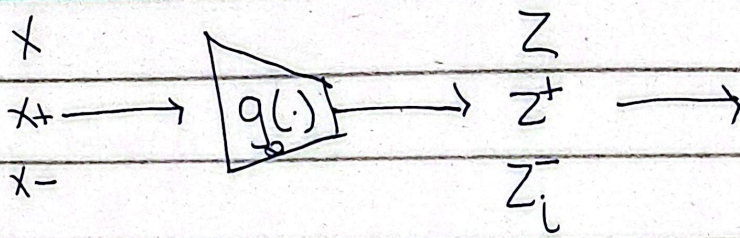
(x, x^+) — Positive pair

(x, x^-) — Negative pair



goal: bring x and x^+ closer

bring x move x and x^- further away



Info NCE loss

$$L_{\text{info NCE}} = - \frac{e^{\sin(x, x^+)}}{\sum_{i=1}^N e^{\sin(x, x_i^-)}}$$

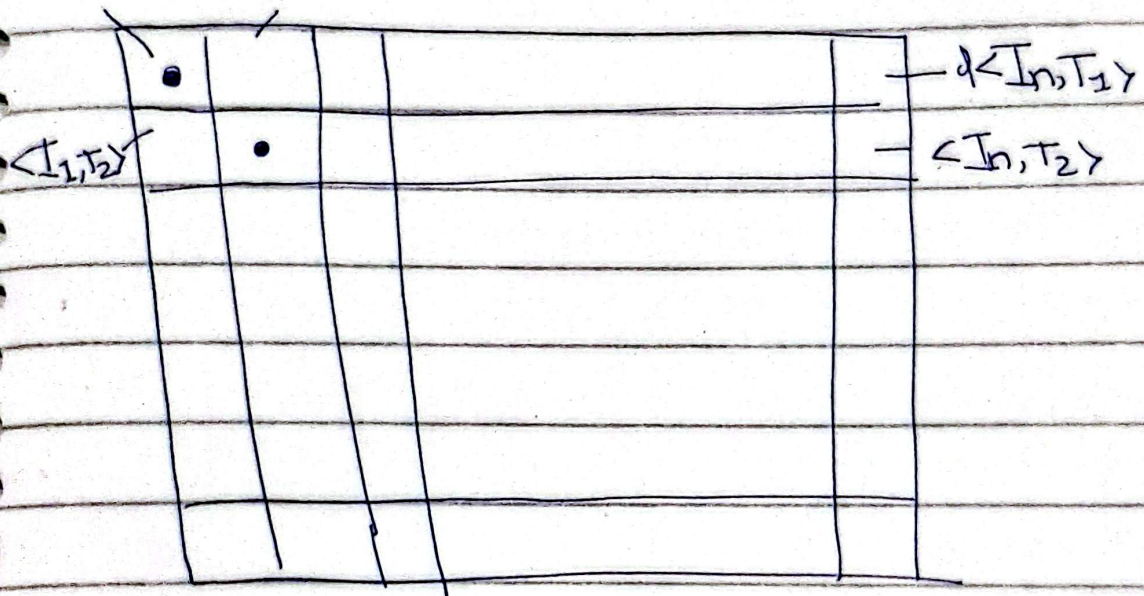
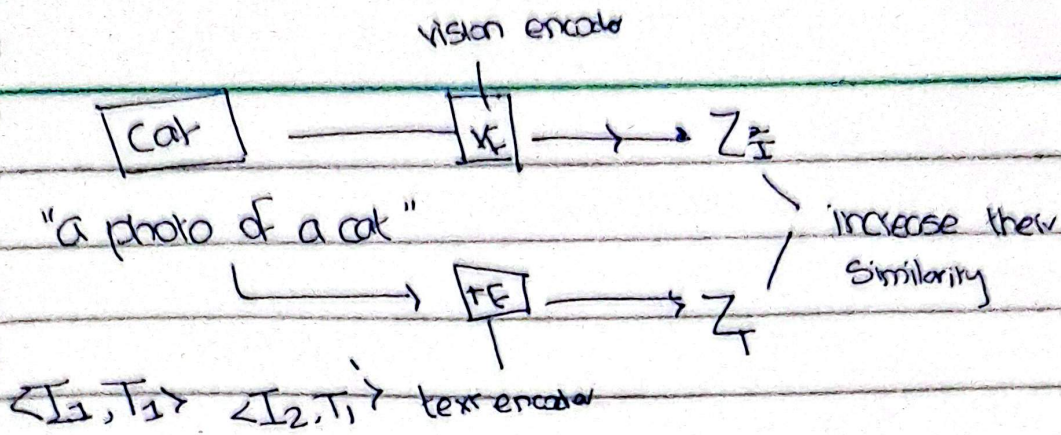
* Now, we create a new batch that the new image to cot is anchor now and the other images are x^-

number of anchors

$$L_{\text{info NCE}} = - \sum_{j=1}^n \frac{e^{\sin(x_i, x^+)}}{\sum_{j=1}^n e^{\sin(x_j, x_i^-)}}$$

$$I(x, z) \geq \log N - L_{\text{info NCE}}$$

↑
increase this
by taking a larger
N



$\Rightarrow$ all the diagonals are positive pairs

$\Rightarrow L_{I2T} = \text{column wise Loss}_{\text{info NCE}}$

$L_{T2I} = \text{row wise Loss}_{\text{info NCE}}$

$$L = \frac{1}{2} (L_{I2T} + L_{T2I})$$

Generative Models

$$P_{data} \sim \text{data} : D = \{x_i\}_{i=1}^N$$

- Images
- Videos
- Action

highly complex
(no way of modelling it)

DNN
models

we take 1 by 1 x_i

give it to NN

$$P_{\phi} \approx P_{data}$$

goal: we want to
make a representation
that resembles P_{data}

$$x_i \sim P_{\phi}$$

Distance between P_{data} & P_{ϕ}

and minimize it with respect to

neural network parameters and

that we get a representation that
resembles P_{data}

$$\min_{\phi} D_{KL} (P_{data}, P_{\phi})$$

$$D_{KL} = E_{P_{data}} [\log P_{data}] - E_{P_{data}} [\log P_{\phi}]$$

Q. How to find $E[P_{data}]$

$$* E_{P_{data}} [\log P_{data}]$$

↓
Entropy of P_{data} $H(P_{data})$

$$* E_{P_{data}} [\log P_{\phi}]$$

$$* D_{KL} \downarrow \Rightarrow E_{P_{data}} [\log P_{\phi}] \uparrow$$

$$\downarrow$$
$$\max_{\phi} \left[\log E_{P_{data}} [\log P_{\phi}(x)] \right]$$

$$\downarrow$$
$$\frac{1}{N} \sum_{i=1}^N \log P_{\phi}(x_i)$$

* we don't know P_{data} , but we have sample of x_i from the data

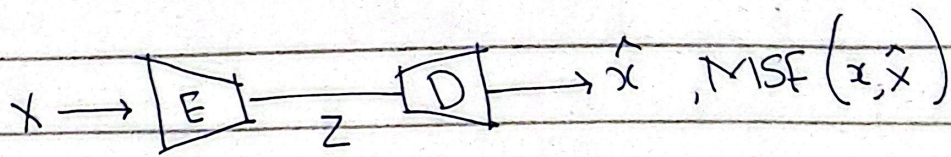
↳ Sample average

→ if $P_{\phi}(x)$ looks more like x_i , then $P_{\phi}(x)$
- a probability should be higher

$$[168, 2] \rightarrow [2, 768]$$

primarily for images

1. Variational Autoencoder:



- Autoencoder's are not generative models

- simple compression models.

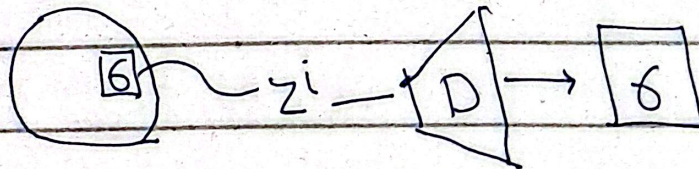
- highly unstructured, we are just minimizing the pixel loss?

- we add noise at 0

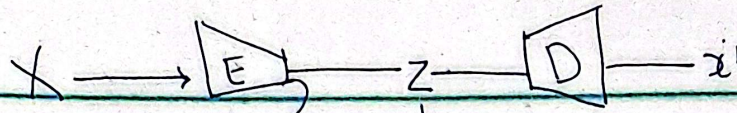
or like adding noise regularization at input

- we need a smooth latent for generation

- Assuming 1 least smooth latent



- if we have holes in the representation, the image becomes unstructured



$$P(z) \sim N(0, I) \quad \text{--- Prior}$$

we assume $\frac{1}{\sqrt{2}}$ is isotropic gaussian distribution

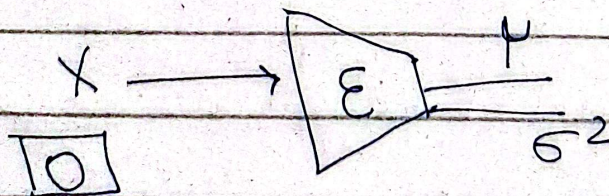
Where does this come from? $z = q_{\psi}(z|x)$ --- Posterior

Contains info abt x

→ we need to bring $q_{\psi}(z|x)$ and $P(z)$ closer

$$D_{KL}(q_{\psi}(z|x), P(z))$$

→ we ask encoder to learn distribution's mean and variance



$$z = q_{\psi}(z|x) \sim N(\mu_e, \sigma_e^2 I)$$

Decoder needs to learn x , so we take

↑
→ Simple MSE error between $x, \bar{x}$

~~MSE~~ MSE =

$$D_{KL}(q_{\psi}(z|x), p(z)) + \underset{p, \theta}{\text{MSE}}(x, \hat{x}) \quad \underset{\text{EBO}}{z} \sim (0, 1)$$

* if $D_{KL}(q_{\psi}(z|x), p(z)) = 0$

↓

means $q_{\psi}(z|x)$ and $p(z)$ is same

↓

no information learnt of x

↓

Posterior Collapse

⇒ Averaging through MSE produces blurry images
because

it ~~is~~ finished sharpness